

DNS (Domain Name System) - Complete Study Notes

I. What is DNS?

Definition

DNS (Domain Name System) is a distributed database that translates human-readable domain names into IP addresses that computers can understand.

Key Analogy

- **Without DNS:** You'd need to type `http://192.78.12.4` instead of `www.google.com`
- **With DNS:** It's like a contact list for the Internet - you know someone's name but need to look up their phone number

How it Works

- Low-level networking only understands raw IP addresses to identify hosts
 - DNS allows humans to use memorable names instead of numerical IP addresses
 - It's a distributed system where different organizations manage their own database portions
 - These databases communicate to build a massive contact list of the Internet
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2. DNS Components

Name Server

- **Purpose:** Loads DNS settings and configs, answers queries from clients or other servers
- **Types:**
 - **Authoritative:** Holds actual DNS records you're looking for
 - **Recursive:** Asks other servers until finding an authoritative server with the records
- **Caching:** Recursive servers can cache information instead of always reaching authoritative servers

Zone File

- **Location:** Lives inside name servers
- **Function:** Stores information about domains or how to reach domains when the server doesn't know directly

Resource Records

- **Structure:** Zone files contain entries of resource records (one per line)
- **Fields:**
 - **Record name:** The domain/hostname
 - **TTL:** Time To Live - how long to keep the record before discarding and getting a new one
 - **Class:** Namespace (commonly "IN" for Internet)
 - **Type:** Type of information (A for address, MX for mail exchanger, etc.)
 - **Data:** Contains IP address (for A records) or other data depending on record type

Example Resource Record

```
patty      IN      A          192.168.0.4
```

3. DNS Process (Resolution Steps)

Example Query: Finding `catzontheinterwebz.com`

Step 1: Local DNS Server

- Host asks: "Where is catzontheinterwebz.com?"
- Local DNS server doesn't know, so it starts from the top
- **Note:** Most users talk to a recursive DNS server provided by their ISPs

Step 2: Root Servers

- **Count:** 13 Root Servers for the Internet (mirrored and distributed worldwide)
- **Function:** Contain information about Top-Level Domains (.org, .com, .net, etc.)
- **Response:** "I don't know catzontheinterwebz.com, but ask the .com TLD server at this IP address"

Step 3: Top-Level Domain (TLD)

- Query sent to .com TLD server

- **Response:** "I don't have catzonthetheinterwebz.com in my zone files, but I have a record for its name server. Here's that IP address."

Step 4: Authoritative DNS Server

- Final request sent to the actual DNS server with the record
- Server has zone file for catzonthetheinterwebz.com
- Finds resource record for 'www' host
- **Response:** Provides the IP address so you can finally see cats on the Internet!

4. Local DNS Configuration

/etc/hosts File

- **Purpose:** Local hostname to IP address mappings
- **Priority:** Checked BEFORE DNS queries
- **Format:** IP address, hostname, aliases

Example /etc/hosts

```
127.0.0.1    localhost
127.0.1.1    icebox
```

Practical Example - Blocking a Site

```
123.45.6.7   www.google.com
```

- Adding this line would redirect Google to a wrong IP address
- Since hosts file is checked first, DNS is never reached
- Result: <www.google.com> becomes inaccessible

/etc/resolv.conf

- **Traditional purpose:** Map DNS name servers for efficient lookups
- **Modern status:** Often irrelevant due to DNS improvements
- **Management:** Usually auto-generated, not manually edited

Example /etc/resolv.conf

```
# conf(5) file for glibc resolver(3) generated by resolvconf(8)
#      DO NOT EDIT THIS FILE BY HAND -- YOUR CHANGES WILL BE OVERWRITTEN
nameserver 127.0.1.1
search localdomain
```

Access Control Files

- **/etc/hosts.deny**: Block access to specific hosts
 - **/etc/hosts.allow**: Allow access to specific hosts
 - **Security Note**: Modifying firewall rules is more secure than using these files
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5. DNS Server Software Options

BIND (Berkeley Internet Name Domain)

- **Status**: Most popular DNS server on the Internet
- **Origin**: Developed at University of California at Berkeley
- **Use case**: Full-featured power and flexibility
- **Best for**: When you need comprehensive DNS functionality

DNSmasq

- **Characteristics**: Lightweight and easy to configure
- **Features**: Includes DHCP and DNS tools
- **Use case**: Simplicity without BIND's complexity
- **Best for**: Smaller networks

PowerDNS

- **Features**: Full-featured, similar to BIND with more flexibility
 - **Database support**: MySQL, PostgreSQL, and others
 - **Advantage**: Easier administration through multiple database options
 - **Philosophy**: Modern alternative to traditional BIND approach
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6. DNS Troubleshooting Tools

nslookup (Name Server Lookup)

- **Purpose:** Query name servers for resource record information
- **Usage:** Find name server information for domains

Example - Finding Google's name server

```
pete@icebox:~$ nslookup www.google.com
Server:           127.0.1.1
Address:          127.0.1.1#53

Non-authoritative answer:
Name:   www.google.com
Address: 216.58.192.4
```

dig (Domain Information Groper)

- **Advantages:** More powerful and flexible than nslookup
- **Use case:** Excellent for troubleshooting DNS issues
- **Output:** Detailed information about DNS queries

Example - Detailed Google DNS query

```
pete@icebox:~$ dig www.google.com

; <<>> DiG 9.9.5-3-Ubuntu <<>> www.google.com
;; global options: +cmd
;; Got answer:
;; ->HEADER<<- opcode: QUERY, status: NOERROR, id: 42376
;; flags: qr rd ra; QUERY: 1, ANSWER: 5, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0005 , udp: 512
;; QUESTION SECTION:
;www.google.com.                IN      A

;; ANSWER SECTION:
www.google.com.                 5       IN      A      74.125.239.147
www.google.com.                 5       IN      A      74.125.239.144
www.google.com.                 5       IN      A      74.125.239.146
www.google.com.                 5       IN      A      74.125.239.145
www.google.com.                 5       IN      A      74.125.239.148

;; Query time: 27 msec
```

```
;; SERVER: 127.0.1.1#53(127.0.1.1)
;; WHEN: Sun Feb 07 10:14:00 PST 2016
;; MSG SIZE rcvd: 123
```

Key Takeaways

1. **DNS is essential** for making the Internet user-friendly by translating names to IP addresses
2. **It's distributed** - no single point of failure, managed by multiple organizations
3. **Resolution is hierarchical** - Root → TLD → Authoritative servers
4. **Local files take precedence** - /etc/hosts is checked before DNS queries
5. **Multiple tools exist** for different needs - from simple DNSmasq to powerful BIND
6. **Troubleshooting tools** like dig and nslookup are essential for DNS diagnosis

Important Note: DNS is an exhaustive topic. This covers basics - additional research is needed for advanced implementation and troubleshooting.